

1 Anomaly Detection by Hand

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Did You Know? John Hopfield and Neural Memory

John Hopfield is a physicist whose work helped connect ideas from physics, biology, and computation. In 1982, he introduced the neural network that would become known as the Hopfield network. Rather than learning to classify inputs in the way many modern neural networks do, a Hopfield network can store patterns and later recover them from incomplete or corrupted versions.

This idea provides an interesting approach to anomaly detection. A Hopfield network can learn patterns representing normal behavior. When a new observation is presented to the network, we can examine whether the network recognizes and reconstructs the observation as one of its stored patterns. Observations that differ substantially from the learned patterns may be considered possible anomalies.

Hopfield's work also illustrates how ideas developed in one scientific field can unexpectedly become useful in another. Concepts inspired by physical systems and associative memory can be applied to problems such as detecting unusual behavior in cyber security data.

In the age of AI, we can increasingly outsource calculations and intellectual tasks to machines, but we cannot outsource our understanding.

1.0.1 Problem 1: From Cyber Security Data to a Hopfield Pattern

A Hopfield network stores and reconstructs patterns. These patterns are commonly represented using bipolar values:

+1 and - 1.

Before a Hopfield network can be used with cyber security data, observations from the system must therefore be converted into a numerical pattern.

Consider a system that monitors user login sessions using the following six features:

Feature	Meaning
f_1	Login from a known device
f_2	Login during normal working hours
f_3	Login from a known location
f_4	Failed authentication occurred
f_5	Connection used an unusual port
f_6	Unusually large data transfer occurred

Assume that the following encoding rule is used:

Yes = +1

and

No = -1.

During one normal login session, the monitoring system records:

Feature	Observation	Value
f_1	Known device	Yes
f_2	Normal working hours	Yes
f_3	Known location	Yes
f_4	Failed authentication	No
f_5	Unusual port	No
f_6	Large data transfer	No

1. Convert each observation into its bipolar value.
2. Construct the Hopfield pattern

$$\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6].$$

3. A second normal login session is observed:

Feature	Observation
f_1	Yes
f_2	No
f_3	Yes
f_4	No
f_5	No
f_6	No

Construct the bipolar pattern for this session.

4. Compare the two patterns. Which feature changed?
5. Explain why it may be useful to store several different patterns of normal behavior rather than defining normal behavior using only one pattern.

1.0.2 Problem 2: Building a Hopfield Weight Matrix by Hand

In the previous problem, cyber security observations were converted into bipolar patterns containing +1 and -1.

A Hopfield network can learn these patterns using a form of Hebbian learning. For a single pattern $\mathbf{x}$, the weight matrix can be constructed from the outer product

$$\mathbf{x}\mathbf{x}^T.$$

Consider a simplified login monitoring system using four features:

Feature	Meaning
f_1	Login from a known device
f_2	Login during normal working hours
f_3	Failed authentication occurred
f_4	Unusual port used

A normal login session produces the following observations:

Feature	Observation
f_1	Yes
f_2	Yes
f_3	No
f_4	No

Use the bipolar encoding

$$\text{Yes} = +1, \quad \text{No} = -1.$$

1. Convert the cyber security observations into a bipolar pattern $\mathbf{x}$.
2. Write the pattern as a column vector.
3. Calculate the outer product

$$\mathbf{x}\mathbf{x}^T.$$

Recall that every element of the resulting matrix is calculated as

$$w_{ij} = x_i x_j.$$

4. Hopfield networks do not allow a neuron to contribute to its own activation. Therefore, set all diagonal elements of the matrix to zero.
The resulting matrix is the Hopfield weight matrix W .
5. What do positive and negative weights mean in this particular stored pattern?

1.0.3 Problem 3: Storing Multiple Normal Cyber Security Patterns

In the previous problem, a Hopfield network stored a single normal pattern. However, normal cyber security behavior may take several forms. A user, for example, may sometimes log in outside normal working hours while still behaving legitimately.

A Hopfield network can store multiple patterns by combining their individual weight matrices.

For P patterns, the weight matrix is calculated as

$$W = \frac{1}{P} \sum_{p=1}^P \mathbf{x}^{(p)} \mathbf{x}^{(p)T}.$$

The diagonal elements are then set to zero.

Consider the same four cyber security features:

Feature	Meaning
f_1	Login from a known device
f_2	Login during normal working hours
f_3	Failed authentication occurred
f_4	Unusual port used

Two different login behaviors have been identified as normal.

Feature	Normal Pattern 1	Normal Pattern 2
f_1	Yes	Yes
f_2	Yes	No
f_3	No	No
f_4	No	No

Use the bipolar encoding

$$\text{Yes} = +1, \quad \text{No} = -1.$$

1. Convert both normal cyber security observations into bipolar vectors $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$.
2. Calculate the first outer product

$$\mathbf{x}^{(1)}\mathbf{x}^{(1)T}.$$

3. Calculate the second outer product

$$\mathbf{x}^{(2)}\mathbf{x}^{(2)T}.$$

4. Add the two outer-product matrices.
5. Set the diagonal elements to zero.
6. Since two patterns are being stored, divide the resulting matrix by

$$P = 2.$$

The result is the Hopfield weight matrix W .

7. Explain why storing multiple normal patterns may be useful for cyber security anomaly detection.

1.0.4 Problem 4: Reconstructing a Corrupted Cyber Security Pattern

A Hopfield network can act as an associative memory. After a pattern has been stored, the network can sometimes reconstruct the original pattern when it is presented with an incomplete or corrupted version.

Recall the normal cyber security pattern

$$\mathbf{x}^{(N)} = [+1, +1, -1, -1],$$

where the features represent

Feature	Meaning
f_1	Login from a known device
f_2	Login during normal working hours
f_3	Failed authentication occurred
f_4	Unusual port used

The Hopfield network trained on this pattern has the weight matrix

$$W = \begin{bmatrix} 0 & 1 & -1 & -1 \\ 1 & 0 & -1 & -1 \\ -1 & -1 & 0 & 1 \\ -1 & -1 & 1 & 0 \end{bmatrix}.$$

Now suppose a new login session is observed:

Feature	Observation
f_1	Yes
f_2	No
f_3	No
f_4	No

Using

$$\text{Yes} = +1, \quad \text{No} = -1,$$

this observation can be represented as a bipolar vector.

The Hopfield network updates the pattern using

$$\mathbf{h} = W\mathbf{x}$$

followed by the activation function

$$x_i^{new} = \text{sign}(h_i),$$

where

$$\text{sign}(h_i) = \begin{cases} +1 & h_i > 0 \\ -1 & h_i < 0 \end{cases}$$

1. Convert the new cyber security observation into a bipolar vector $\mathbf{x}$.
2. Compare the new observation with the stored normal pattern. Which feature is different?
3. Calculate

$$\mathbf{h} = W\mathbf{x}$$

by hand.

4. Apply the sign function to each element of $\mathbf{h}$ to obtain

$$\mathbf{x}^{new}$$

5. Compare the reconstructed pattern with the original stored normal pattern.
6. Explain what the reconstruction means in the context of this cyber security example.

1.0.5 Problem 5: Using Hopfield Reconstruction for Anomaly Detection

In the previous problem, a Hopfield network reconstructed a corrupted cyber security observation as a stored normal pattern.

This reconstruction behavior can also be used for anomaly detection.

One simple approach is to compare an observed pattern with the pattern reconstructed by the Hopfield network. If the observation is close to a stored normal pattern, only a small number of features may need to change during reconstruction. If the observation is substantially different from the learned normal behavior, more features may need to change.

A simple measure of this difference is the Hamming distance.

For two bipolar vectors $\mathbf{x}$ and $\hat{\mathbf{x}}$,

$$d_H(\mathbf{x}, \hat{\mathbf{x}}) = \sum_{i=1}^n I(x_i \neq \hat{x}_i),$$

where

$$I(x_i \neq \hat{x}_i) = \begin{cases} 1 & x_i \neq \hat{x}_i \\ 0 & x_i = \hat{x}_i. \end{cases}$$

In other words, the Hamming distance simply counts the number of positions where the two vectors are different.

Consider a Hopfield network that has learned the normal cyber security pattern

$$\mathbf{x}^{(N)} = [+1, +1, -1, -1].$$

The four features represent

Feature	Meaning
f_1	Login from a known device
f_2	Login during normal working hours
f_3	Failed authentication occurred
f_4	Unusual port used

The trained weight matrix is

$$W = \begin{bmatrix} 0 & 1 & -1 & -1 \\ 1 & 0 & -1 & -1 \\ -1 & -1 & 0 & 1 \\ -1 & -1 & 1 & 0 \end{bmatrix}.$$

Two new login sessions are observed.

$$\mathbf{x}^{(A)} = [+1, -1, -1, -1]$$

and

$$\mathbf{x}^{(B)} = [-1, -1, +1, +1].$$

1. For observation A, calculate

$$W\mathbf{x}^{(A)}$$

and apply the sign function to obtain the reconstructed pattern

$$\hat{\mathbf{x}}^{(A)}.$$

2. Calculate

$$d_H(\mathbf{x}^{(A)}, \hat{\mathbf{x}}^{(A)}).$$

3. Repeat the reconstruction for observation B to obtain

$$\hat{\mathbf{x}}^{(B)}.$$

4. Calculate

$$d_H \left(\mathbf{x}^{(B)}, \hat{\mathbf{x}}^{(B)} \right).$$

5. Which observation required the largest change during reconstruction?
6. Based on the cyber security features, explain why observation B may deserve investigation by a security analyst.
7. Explain why a large reconstruction difference should be considered an anomaly signal rather than proof of an attack.

1.0.6 Problem 6: Hopfield Energy and Attractors

A Hopfield network can also be understood as an energy-based system. Every state of the network has an associated energy.

For a Hopfield network with state vector $\mathbf{x}$ and weight matrix W , the energy is

$$E(\mathbf{x}) = -\frac{1}{2} \mathbf{x}^T W \mathbf{x}.$$

Stable stored patterns tend to correspond to low-energy states of the network. These stable states are sometimes called *attractors* because nearby patterns may move toward them as the network updates.

Consider the Hopfield network from the previous problems:

$$W = \begin{bmatrix} 0 & 1 & -1 & -1 \\ 1 & 0 & -1 & -1 \\ -1 & -1 & 0 & 1 \\ -1 & -1 & 1 & 0 \end{bmatrix}.$$

The stored normal cyber security pattern is

$$\mathbf{x}^{(N)} = [+1, +1, -1, -1].$$

Now consider the following observed pattern:

$$\mathbf{x}^{(0)} = [+1, -1, -1, -1].$$

This observation differs from the stored normal pattern in one feature.

Asynchronous Updating

In this problem, the neurons will be updated one at a time. This is called an asynchronous update.

For neuron i , calculate

$$h_i = \sum_j w_{ij} x_j$$

and then update only that neuron using

$$x_i^{new} = \text{sign}(h_i).$$

For a symmetric Hopfield weight matrix with a zero diagonal, asynchronous updates provide an important property: the network energy cannot increase when a neuron is updated according to the Hopfield rule.

1. Calculate

$$W\mathbf{x}^{(0)}.$$

2. Calculate the initial energy

$$E(\mathbf{x}^{(0)}) = -\frac{1}{2}\mathbf{x}^{(0)T}W\mathbf{x}^{(0)}.$$

3. Examine the activation of neuron 2. Determine its updated value using

$$x_2^{new} = \text{sign}(h_2).$$

Update only neuron 2 and write the resulting state $\mathbf{x}^{(1)}$.

4. Calculate the energy

$$E(\mathbf{x}^{(1)}).$$

5. Compare the energy before and after the update.
6. Compare $\mathbf{x}^{(1)}$ with the stored normal pattern $\mathbf{x}^{(N)}$.
7. Explain what an attractor means in the context of a Hopfield network being used to represent normal cyber security behavior.

1.0.7 Problem 7: How Many Patterns Can a Hopfield Network Store?

In the previous problems, a Hopfield network was used to store patterns of normal cyber security behavior. This raises an important question:

How many different patterns can a Hopfield network reliably store?

A Hopfield network does not have unlimited memory. As more patterns are stored, the patterns begin to interfere with each other and reliable reconstruction becomes more difficult.

For a classical Hopfield network containing d neurons, a commonly used approximation for its storage capacity is

$$C \approx 0.138d,$$

where C is the approximate number of random bipolar patterns that can be stored before retrieval errors become significant.

Another, more conservative approximation associated with high-probability reliable retrieval is

$$C \approx \frac{d}{2 \ln(d)}.$$

These equations describe different notions of storage capacity and should be viewed as approximate guidelines rather than exact limits.

Cyber Security Scenario

Suppose a cyber security system represents each network session using a bipolar feature vector containing

$$d = 100$$

features.

Each stored pattern represents one form of normal network behavior.

1. Using

$$C \approx 0.138d,$$

estimate the number of patterns that the Hopfield network can store.

2. Using

$$C \approx \frac{d}{2 \ln(d)},$$

calculate the more conservative capacity estimate.
For this calculation, use

$$\ln(100) \approx 4.605.$$

3. Suppose the security analyst attempts to store 30 different normal behavior patterns.
Compare this number with the two capacity estimates.
4. Now suppose the system is redesigned to use

$$d = 1000$$

features.
Using

$$C \approx 0.138d,$$

estimate the new storage capacity.

5. Explain why increasing the number of features can increase the storage capacity of a Hopfield network.
6. Explain why a cyber security system should not simply attempt to store every normal network observation as a separate Hopfield pattern.

1.0.8 Problem 8: Building a Hopfield Cyber Security Anomaly Detector

In the previous problems, we examined the individual steps required to use a Hopfield network for anomaly detection. We will now combine these steps into a complete example.

Consider a cyber security monitoring system using four features:

Feature	Meaning
f_1	Login from a known device
f_2	Login during normal working hours
f_3	Failed authentication occurred
f_4	Unusual port used

The system uses the bipolar encoding

$$\text{Yes} = +1, \quad \text{No} = -1.$$

During a period of known normal operation, the following two types of login behavior are observed:

Feature	Normal A	Normal B
Known device	Yes	Yes
Normal working hours	Yes	No
Failed authentication	No	No
Unusual port	No	No

A Hopfield network will be trained using these two normal patterns.

Recall that the Hopfield weight matrix is

$$W = \frac{1}{P} \sum_{p=1}^P \mathbf{x}^{(p)} \mathbf{x}^{(p)T},$$

with the diagonal elements set to zero.

After training, a new login session is observed:

Feature	New Observation
Known device	No
Normal working hours	No
Failed authentication	Yes
Unusual port	No

1. Convert Normal A and Normal B into bipolar vectors $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$.
2. Calculate

$$\mathbf{x}^{(1)}\mathbf{x}^{(1)T}$$

and

$$\mathbf{x}^{(2)}\mathbf{x}^{(2)T}.$$

3. Add the matrices, set the diagonal elements to zero, and divide by $P = 2$ to obtain the Hopfield weight matrix W .
4. Convert the new login observation into a bipolar vector $\mathbf{x}$.
5. Calculate

$$W\mathbf{x}.$$

6. Apply the sign function to obtain the reconstructed pattern

$$\hat{\mathbf{x}} = \text{sign}(W\mathbf{x}).$$

For this problem, if an activation is exactly zero, leave the corresponding feature unchanged.

7. Calculate the Hamming distance between the new observation and each stored normal pattern:

$$d_H(\mathbf{x}, \mathbf{x}^{(1)})$$

and

$$d_H(\mathbf{x}, \mathbf{x}^{(2)}).$$

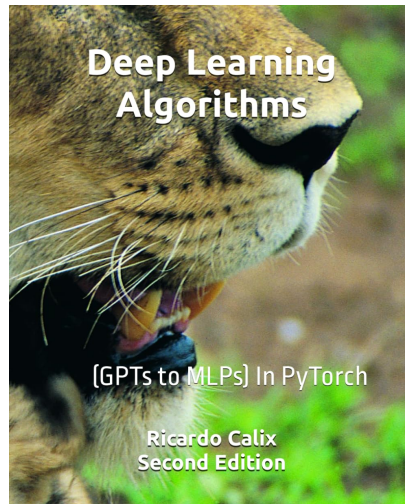
8. Calculate the Hamming distance between the reconstructed pattern and each stored normal pattern.
9. Based on the reconstruction and the Hamming distances, would this observation deserve investigation as a possible anomaly? Explain your answer using the cyber security features.

10. Explain why the result should be considered an anomaly signal rather than proof that an attack occurred.

1.1 Summary

Mastering these problems is fundamental to mastering Anomaly Detection for cyber security.

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